Procurement & Vendor Spend Analytics Pipeline

Data Quality Summary

Project Objective

Build an end-to-end procurement analytics pipeline using Databricks, PySpark, Spark SQL and Delta Lake with a Bronze–Silver–Gold architecture.

1. Bronze Layer – Raw Ingestion

Raw procurement data was loaded into Delta tables while preserving source records. Metadata columns including ingestion_timestamp and source_file were added.

2. Silver Layer – Data Cleaning

The Silver layer reads Bronze tables and performs basic data-quality processing. Critical null records are handled, duplicates are removed, timestamps and numeric fields are cast to appropriate types, and text values are standardized.

3. SCD Type 2 – Vendor Contracts

Vendor contract history is maintained using Delta MERGE logic. When the negotiated price changes, the previous active record is marked inactive with an end date and the new version is inserted as active.

4. Gold Layer – Business Analytics

Business-ready tables are created for vendor spend, price variance, vendor risk classification, and regional spend analysis. Positive price variance is used to identify overcharged records.

5. Validation Performed

Schema and result checks were performed after major processing steps. Duplicate checks and historical SCD2 records were reviewed before the Gold layer.

6. Data Quality Outcome

The pipeline separates raw, cleaned and analytical data, preserves historical contract versions, and produces structured Gold tables suitable for reporting and dashboarding.

Technology Stack

Python | PySpark | Spark SQL | Delta Lake | Databricks | ADLS Gen2